Arturo Zhang Luo

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Education

Simon Fraser University Vancouver, BC

BSc. in Computing Science *Sep. 2024 – Aug. 2028*

- Relevant Coursework: Data Structures & Algorithms, Introduction to Software Engineering, Artificial Intelligence, Operating Systems, Systems Programming

Experience

E-Commerce & Technical Support Assistant Sep. 2022 – Aug. 2023

Distribuidora Supermarket Trolley *Panama*

- Advised clients on product selection for furniture and mattresses, consistently closing sales through consultative communication and product knowledge.
- Maintained accurate **product listings on the company's online store**, ensuring descriptions, pricing, and inventory availability remained current across all categories.
- Led the design and implementation of a **barcode tagging system** for over 200 products, streamlining **inventory management** and improving checkout efficiency across departments.

Projects

aTour — AI Itinerary Builder | *Python, FastAPI, PostgreSQL, React Native, Gemini API* May 2026 – Present

- Building a **two-sided marketplace** connecting local tour guides and tourists, with guides managing up to **5 experience listings** and tourists discovering them via **natural language AI matching**.
- Designed a normalized **PostgreSQL schema** (7 tables: User, Post, Slot, Booking, Review) with **JWT + OAuth** authentication, eliminating unauthorized access in all testing scenarios.
- Integrated **Google Gemini API** to semantically rank listings against tourist queries, reducing average search-to-match time by ∼**60%** compared to manual browse-only flow.
- Containerized the full stack with **Docker Compose**, enabling one-command local setup and cutting new contributor onboarding time by **over 80%**.

Multithreaded Group Chat Server & Client | *C, POSIX Threads, TCP Sockets* April 2026

- Engineered a concurrent **TCP group chat server** in C supporting up to **100 simultaneous clients**, achieving **zero message reordering** across all connections under load testing.
- Designed a compact **binary messaging protocol** with typed frames, reducing per-message parsing overhead by ∼**40%** compared to a text-based equivalent.
- Implemented a **Two-Phase Commit shutdown protocol** using POSIX mutexes, eliminating race conditions and reducing abnormal termination errors by **100%**.

Movie Trivia | *Python, FastAPI, PostgreSQL, SQLAlchemy, JavaScript, HTML/CSS* Sep. 2025 – Dec. 2025

- Developed a **full-stack web trivia game** in an Agile team of 5, integrating **JWT authentication (RS256)** and a tiered rate-limiting system that cut API abuse attempts by **over 90%**.
- Built a **PST-aware daily challenge system** enforcing one attempt per user per day, alongside a leaderboard with time-filtered queries (today / this week / all time).
- Deployed to **Render** with **99%+ uptime**; shipped a friend system with **4 interaction states** and persistent user profiles with TMDB movie search integration.

Certifications

Machine Learning Specialization | *Stanford University & DeepLearning.AI – Coursera* August 2025

- Trained ML models with NumPy, scikit-learn, and TensorFlow; applied supervised and unsupervised learning, recommender systems, and reinforcement learning to real-world datasets.

Technical Skills

Programming & Scripting: Python, C, C++, JavaScript, TypeScript, HTML, CSS
Frameworks & Libraries: FastAPI, React.js, React Native, SQLAlchemy, Bootstrap
Tools & Platforms: Git/GitHub, Docker, Docker Compose, PostgreSQL, Linux, VSCode, Render, Vercel
Practices: Agile Methodologies, CI/CD, RESTful API Design, Version Control